WORKED SOLUTION

STEP III 2008, Question 6

Pure • First-Order Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q6

The key idea

Each equation gives y directly in terms of x and the slope p, but it is not solved for p, so we can't simply integrate. Differentiating with respect to x removes y (because $\mathrm{d}y/\mathrm{d}x = p$) and leaves a relation between x, p and $\mathrm{d}p/\mathrm{d}x$. Turned upside down, that relation is a *linear* first-order equation for x as a function of p, which an integrating factor solves. Then we put $x(p)$ back into the original equation to get y.

Solution

Part (i)

Differentiate $y = p^2 + 2xp$ with respect to x, using the chain rule on p^2 and the product rule on $2xp$, and replace $\mathrm{d}y/\mathrm{d}x$ by p:

$$p = 2p\frac{\mathrm{d}p}{\mathrm{d}x} + 2p + 2x\frac{\mathrm{d}p}{\mathrm{d}x} \quad \Longrightarrow \quad -p = (2p + 2x)\frac{\mathrm{d}p}{\mathrm{d}x}.$$

? **Ask yourself** Are we allowed to flip $\mathrm{d}p/\mathrm{d}x$ over to get $\mathrm{d}x/\mathrm{d}p$?

We need $\mathrm{d}p/\mathrm{d}x \neq 0$. The equation above shows that wherever $p \neq 0$, both $\mathrm{d}p/\mathrm{d}x$ and $2p + 2x$ are non-zero. (We divide by p anyway, so $p \neq 0$ is needed throughout.) So we may write

$$\frac{\mathrm{d}x}{\mathrm{d}p} = -\frac{2p + 2x}{p} = -2 - \frac{2x}{p},$$

as required.

Rearranged, this is linear in x:

$$\frac{\mathrm{d}x}{\mathrm{d}p} + \frac{2}{p}x = -2.$$

The integrating factor is $e^{\int 2/p\,dp} = p^2$, and multiplying through gives

$$\frac{d}{dp}(p^2x) = -2p^2 \quad \Longrightarrow \quad p^2x = -\tfrac{2}{3}p^3 + A \quad \Longrightarrow \quad x = -\tfrac{2}{3}p + Ap^{-2},$$

with A an arbitrary constant. That is the stated result.

Now use $p = -3$ when $x = 2$:

$$2 = -\tfrac{2}{3}(-3) + \frac{A}{9} = 2 + \frac{A}{9},$$

so $A = 0$ and $x = -\frac{2}{3}p$, that is, $p = -\frac{3}{2}x$. Substituting into the original equation,

$$y = p^2 + 2xp = \tfrac{9}{4}x^2 + 2x\left(-\tfrac{3}{2}x\right) = \tfrac{9}{4}x^2 - 3x^2,$$

so

$$y = -\tfrac{3}{4}x^2.$$

As a check, $dy/dx = -\frac{3}{2}x$, which really is p, and it equals -3 at $x = 2$.

Part (ii)

Here $\ln p$ appears, so $p > 0$. Differentiate $y = 2xp + p\ln p$ with respect to x, using $\frac{d}{dp}(p\ln p) = \ln p + 1$:

$$p = 2p + 2x\frac{dp}{dx} + (\ln p + 1)\frac{dp}{dx} \quad \Longrightarrow \quad -p = (2x + \ln p + 1)\frac{dp}{dx}.$$

As before, $p \neq 0$ forces $dp/dx \neq 0$, so

$$\frac{dx}{dp} = -\frac{2x + \ln p + 1}{p} \quad \Longrightarrow \quad \frac{dx}{dp} + \frac{2}{p}x = -\frac{\ln p + 1}{p}.$$

The integrating factor is again p^2:

$$\frac{d}{dp}(p^2x) = -p\ln p - p.$$

Integrating by parts with $u = \ln p$, $dv = p\,dp$,

$$\int p\ln p\,dp = \tfrac{1}{2}p^2\ln p - \int \tfrac{1}{2}p\,dp = \tfrac{1}{2}p^2\ln p - \tfrac{1}{4}p^2,$$

so

$$\begin{aligned} p^2x &= -\tfrac{1}{2}p^2\ln p + \tfrac{1}{4}p^2 - \tfrac{1}{2}p^2 + B \\ &= -\tfrac{1}{2}p^2\ln p - \tfrac{1}{4}p^2 + B, \end{aligned}$$

and

$$x = -\tfrac{1}{2}\ln p - \tfrac{1}{4} + Bp^{-2}.$$

Using $p = 1$ when $x = -\frac{1}{4}$: $-\frac{1}{4} = 0 - \frac{1}{4} + B$, so $B = 0$ and

$$x = -\tfrac{1}{2}\ln p - \tfrac{1}{4},$$

which is the result we were asked to show.

To find y, notice that this says $\ln p = -2x - \frac{1}{2}$. Substituting this into the original equation is quicker than substituting for p everywhere:

$$y = 2xp + p\left(-2x - \tfrac{1}{2}\right) = -\tfrac{1}{2}p,$$

and $p = \mathrm{e}^{-2x-1/2}$, so

$$y = -\tfrac{1}{2}\,\mathrm{e}^{-2x-\frac{1}{2}}.$$

Check: $\mathrm{d}y/\mathrm{d}x = \mathrm{e}^{-2x-\frac{1}{2}} = p$, as it must be, and at $x = -\frac{1}{4}$ this is $\mathrm{e}^0 = 1$.

Checks

In each part, differentiate your final y and confirm that the derivative really is the p you used: $y = -\frac{3}{4}x^2$ gives $p = -\frac{3}{2}x$, and $p^2 + 2xp = \frac{9}{4}x^2 - 3x^2 = y$. For part (ii) at $x = 0$, $p = \mathrm{e}^{-1/2} \approx 0.6065$ and $2xp + p\ln p = 0.6065 \times (-0.5) \approx -0.3033 = y(0)$. A numerical check also confirmed that, for random values of A and B, the general parametric curves $(x(p), y(p))$ have slope $\mathrm{d}y/\mathrm{d}x$ equal to p.

Common slips

- Forgetting that $\mathrm{d}y/\mathrm{d}x$ on the left-hand side is p, and so failing to see the cancellation that leaves $-p$.
- Dropping the constant of integration before using the initial condition. Here it happens to vanish, but you only know that after finding it.
- In part (ii), differentiating $p\ln p$ with respect to x as $\ln p$ alone, forgetting the factor $\mathrm{d}p/\mathrm{d}x$ and the extra $+1$.

Beyond STEP

Equations of the form $y = xf(p) + g(p)$ are called d'Alembert (or Lagrange) equations; the special case $y = xp + g(p)$ is Clairaut's equation. The trick of treating p as the independent variable works for all of them. They can also have *singular solutions*, which arise where we divided by $\mathrm{d}p/\mathrm{d}x$ and which are envelopes of the family of general solutions. Boyce and DiPrima, *Elementary Differential Equations*, covers first-order equations and their methods of solution.

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